Unsupervised Learning and Dimensionality Reduction Report

Syed Muhammad Husainie
Georgia Institute of Technology
GT ID: 903965504
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***Abstract*—For this study, unsupervised learning techniques including clustering and dimensionality reduction were implemented to understand the downstream influence on neural network performance on the Adult Income and Wine Quality datasets. First, Gaussian Mixture Models (GMM) and K means clustering were applied to identify natural groupings. Then, Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Random Projection were evaluated to compress feature spaces and preserve meaningful structures. Finally, the best neural network architecture from prior supervised learning research was retrained on the reduced feature space to measure changes in predictive performance. The findings suggest that while dimensionality reduction can enhance cluster interpretability, the overall impact on neural network models is nuanced.**

I. Introduction

This study investigates how unsupervised learning techniques reveal underlying structure in data and influence neural network performance. The goal is to determine whether transformations that improve runtime can improve downstream classification performance, convergence, or feature efficiency without modifying the baseline architecture. Experiments are conducted on the Adult Income (binary) and Wine Quality (multiclass) datasets.

Following past research approaches, unsupervised methods for this study are first used to explore and capture latent structure in the data, followed by transformations that compress the feature space while aiming to preserve meaningful information. These representations are then analyzed to assess how well they maintain or improve data structure. Finally, the same neural network architecture implemented in prior work is retrained on these transformed features, and additional representations derived from clustering are explored to evaluate their impact on classification performance.

II. Datasets and Exploratory Data Analysis

An exploratory data analysis was carried out for both datasets to help guide clustering and dimensionality reduction hypotheses on the Adult Income and Wine Quality datasets. These datasets were previously used in the supervised learning report and therefore maintain the same preprocessing pipeline and train–validation–test splits to ensure comparability across experiments.

A. Adult Income Dataset

The Adult Income dataset involves predicting whether an individual's annual income exceeds $50,000 based on demographic and employment-related attributes. The dataset contains 45,175 observations after removing duplicate rows, with a total of 15 features including both categorical and numerical attributes.

Several important characteristics are noted from the exploratory analysis. First, the dataset contains a mix of categorical and continuous variables, requiring encoding and normalization before applying clustering and dimensionality reduction algorithms. Second, several numerical variables such as *capital-gain* and *capital-loss* have heavy right skew due to a large number of zero values and a small number of extreme observations. Third, the target variable has moderate class imbalance, with approximately 75% of instances belonging to the $\leq$50K class and 25% belonging to the $>$50K class.

These properties make the Adult dataset a useful benchmark for studying clustering and dimensionality reduction under data imbalance. In particular, the presence of categorical variables may affect distance-based clustering algorithms such as K-Means, while skewed distributions influence interpretability in PCA and ICA.

B. Wine Quality Dataset

The Wine Quality dataset focuses on predicting wine quality scores based on physicochemical measurements of wine samples. This task represents a multiclass classification problem with continuous input features and moderate label imbalance across quality categories. The dataset consists of 5,320 unique observations after removing duplicate rows and includes 14 numerical attributes describing characteristic properties of wine samples, with quality ratings scale ranging from 1 to 8.

EDA reveals several important properties of this dataset. Most features are continuous chemical measurements with moderate right skew, particularly variables such as residual sugar and chlorides. The target variable is ordinal and exhibits class imbalance, with the majority of samples clustered around mid-range quality values (3–5) and relatively few observations in extreme quality categories. The dataset introduces additional complexity due to an overlap of class boundaries and measurement noise.

Compared to the Adult dataset, Wine Quality contains only numerical inputs but shows greater overlap between classes and noisier feature relationships. These characteristics create a more challenging platform for clustering and dimensionality reduction, making the dataset a useful test basis for evaluating whether these techniques can reveal meaningful structure despite noise.

III. Hypothesis

Combining dimensionality reduction with clustering is expected to produce more interpretable structure than applying clustering directly on the original feature space. In addition, augmenting the original inputs with cluster derived features may provide complementary information that improves neural network classification performance.

These hypotheses are grounded in established ideas from unsupervised learning. Dimensionality reduction methods such as PCA and ICA aim to remove noise and highlight latent structure that may better align with natural groupings [1], [2]. Clustering methods are designed to capture these groupings, with GMM offering soft probabilistic assignments that may retain more information than the hard assignments produced by K-Means [3]. When these cluster-based representations are incorporated alongside the original features, they may act as additional signals that help improve generalization.

Based on the characteristics of each dataset, the following expectations emerge.

a) Adult Dataset: The presence of categorical variables, data imbalance, and skewed numerical features may make distance-based methods like K-Means sensitive to preprocessing choices such as scaling and encoding. GMM may be more flexible in handling this structure through probabilistic assignments. Furthermore, dimensionality reduction is expected to help by mapping the data into a space where distance-based assumptions are more meaningful. For classification, cluster derived features are expected to provide modest gains by capturing relationships that are not fully represented in the original feature set.

b) Wine Dataset: The Wine dataset consists of continuous features with overlapping class boundaries and some measurement noise. Clustering may still uncover partial structure related to quality levels, but alignment with labels may be imperfect. ICA may help separate underlying sources of variation that reflect different chemical properties [2]. However, the noisier structure may limit the benefit of cluster derived features, as the resulting groupings may not correspond well enough to the target labels.

IV. Methods

A. Preprocessing

All features were standardized using `StandardScaler` to ensure equal contribution to distance based methods. For the Adult dataset, categorical variables were one hot encoded, while the Wine dataset used only numerical features with `wine type` included. Data was split 80/20 using stratified sampling, and all unsupervised models were fit only on training data to avoid leakage.

B. Clustering Algorithms

K-Means and GMMs were evaluated. For K-Means, the number of clusters $k \in [2, 10]$ was selected using silhouette score with multiple initializations. For GMM, the number of components was chosen by minimizing BIC using full covariance matrices. Stability was assessed using Adjusted Rand Index across random seed.

C. Dimensionality Reduction

PCA, ICA, and Random Projection were applied. ICA components were selected based on maximizing kurtosis, while Random Projection dimensionality was chosen using reconstruction error.

D. Neural Network Architecture

A two-layer network with 64 hidden units, batch normalization, ReLU, and 0.3 dropout was used. Training used Adam ($\beta_1 = 0$, $\beta_2 = 0.999$), learning rate 10^{-3}, and weight decay 10^{-4}. Cross-entropy with label smoothing (0.05), early stopping, and batch size 256 were applied. All experiments used identical settings across feature variants to ensure fair comparison.

V. Results

A. Step 1: Clustering on Raw Data

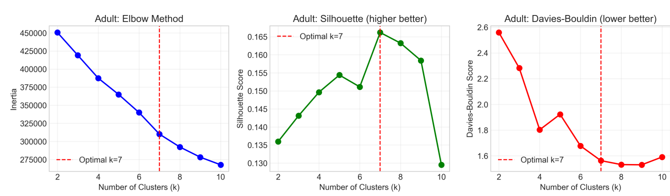

Fig. 1. K-Means clustering evaluation metrics for the Adult dataset. Left: Elbow method showing inertia versus number of clusters. Center: Silhouette scores indicating cluster quality. Right: Davies-Bouldin scores where lower values indicate better separation. Optimal cluster count $k = 7$ is selected at the silhouette peak.

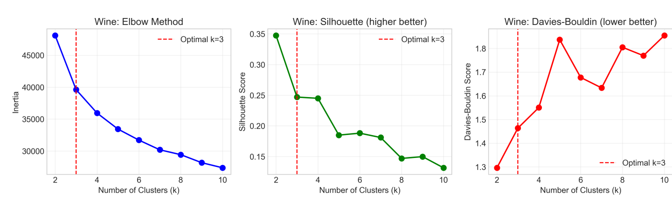

Fig. 2. K-Means clustering evaluation metrics for the Wine dataset. Left: Elbow method showing inertia decreasing with increasing clusters. Center: Silhouette scores peaking at $k = 3$ with value 0.247. Right: Davies-Bouldin scores showing minimum at $k = 3$, confirming optimal cluster selection.

Clustering algorithms were applied to the raw feature spaces of both datasets to identify natural groupings without using ground truth labels. Optimal hyperparameters were selected using label free criteria including silhouette score maximization for K-Means and Bayesian Information Criterion minimization for Gaussian Mixture Models.

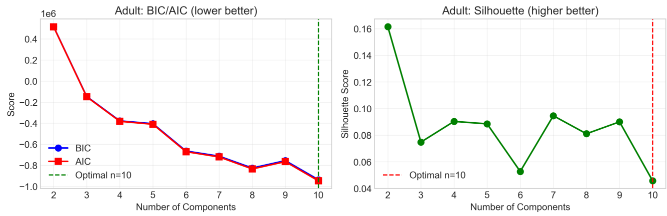

Fig. 3. Gaussian Mixture Model evaluation for the Adult dataset. (a) BIC and AIC scores across number of components, with BIC minimization selecting $n = 10$ components. (b) Silhouette scores showing decreasing trend, indicating that additional components add complexity without improving cluster cohesion.

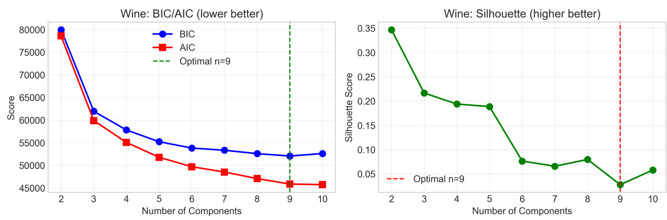

Fig. 4. Gaussian Mixture Model evaluation for the Wine dataset. (a) BIC and AIC curves showing minimization at $n = 9$ components. (b) Silhouette scores across components, with peak values substantially lower than K-Means, indicating that GMM assumptions likely do not align well with the underlying data structure.

1) K-Means Clustering: For the Adult dataset, silhouette scores were evaluated for $k \in [2, 10]$, with the optimal number of clusters selected at $k = 7$ achieving a silhouette score of 0.166. The elbow curve showed decreasing inertia with increasing k, while silhouette scores were at their highest at $k = 7$ before declining, which indicate that seven clusters provide the best balance between clusters. For the Wine dataset, the optimal number of clusters was $k = 3$, with a silhouette score of 0.247. The higher silhouette score for Wine suggests that the dataset exhibits stronger natural cluster structure compared to Adult, likely due to its continuous feature space and intrinsic relationships among wine samples.

Post analysis stability was measured using Adjusted Rand Index across five different initializations. K-Means was fairly stable on Adult (mean ARI = 0.767) but completely stable on Wine (mean ARI = 1.000). The lower stability on Adult is likely due to the mix of categorical and numerical features, along with class imbalance, which makes the clustering more sensitive to initialization.

To understand how meaningful the clusters are, they were compared to the true labels using ARI and NMI for analysis only (Table 1). For Adult, the ARI of 0.033 shows that the clusters do not align with income labels at all. For Wine, the ARI of 0.157 indicates some weak alignment, meaning the clusters capture a small amount of structure related to quality, but not strongly.

2) Gaussian Mixture Models: GMMs with full covariance matrices were evaluated for $n \in [2, 10]$ components. For the Adult dataset, BIC minimization selected $n = 10$ components, with silhouette score of 0.046. The BIC curve showed a notably decreasing trend up to $n = 10$, suggesting that additional components continue to improve model fit despite diminishing returns in silhouette quality. For the Wine dataset, BIC minimization selected $n = 9$ components, achieving a silhouette score of 0.028, substantially lower than K Means performance on the same dataset.

Stability analysis revealed that GMM clustering is less stable than K-Means across both datasets, with mean ARI of 0.657 for Adult and 0.584 for Wine. This reduced stability can be attributed to the increased flexibility of GMMs, which are more susceptible to initialization sensitivity and local optima in the expectation-maximization procedure.

Alignment with ground truth labels was slightly higher for GMM compared to K-Means on Adult (ARI $= 0.047$ vs 0.033) but lower on Wine (ARI $= 0.106$ vs 0.157). This suggests that GMM's soft assignments may capture different aspects of the data structure than K-Means hard assignments, with performance varying by dataset characteristics.

TABLE I
STEP 1 SUMMARY: OPTIMAL CLUSTERING RESULTS

Dataset	Algorithm	Optimal	Silhouette	ARI
Adult	K-Means	$k = 7$	0.166	0.033
Adult	GMM	$n = 10$	0.046	0.047
Wine	K-Means	$k = 3$	0.247	0.157
Wine	GMM	$n = 9$	0.028	0.106

3) Summary and Interpretation: Several key observations emerge from the clustering analysis. First, K-Means consistently outperformed GMM in terms of silhouette score and Davies-Bouldin index across both datasets, suggesting that the simpler assumption of spherical clusters with equal variance is more appropriate for these data than the flexible covariance structures offered by GMM. Second, the Wine dataset exhibits stronger cluster structure than Adult, as evidenced by higher silhouette scores and better alignment with ground truth labels, likely due to its continuous feature space and underlying relationships. Third, the optimal number of clusters for Adult ($k = 7$) exceeds the number of classes (binary), indicating that clusters capture fine grained demographic groupings rather than the income distinction directly. Fourth, GMM's lower stability and silhouette scores suggest that the expectation maximization algorithm may struggle with convergence to globally optimal solutions in these feature spaces, specifically when the true underlying distributions deviate from Gaussian assumptions.

These findings support the hypothesis that clustering can reveal meaningful structure in both datasets, though the interpretability and quality of discovered clusters vary substantially by algorithm and dataset characteristics. The superior performance of K Means on both datasets suggests that Euclidean distance is a an effective metric for preprocessed feature spaces, despite the presence of categorical variables in the Adult dataset after encoding.

B. Step 2: Dimensionality Reduction

Dimensionality reduction methods were applied to both datasets to compress feature spaces while preserving meaningful structure. Component counts were selected using label free criteria including explained variance threshold for PCA, kurtosis maximization for ICA, and reconstruction error elbow for Random Projection.

1) Principal Component Analysis: PCA was applied to both datasets with components selected to explain 95% of cumulative variance. For the Adult dataset, this required 13 components, explaining 97.25% of total variance, as indicated in the scree plot in Figure 5 shows diminishing marginal variance beyond the first few components, with a gradual elbow rather than a sharp drop, indicating that Adult features contribute relatively independently to overall variance. Analyzing the component loadings reveal that PC1 captures relationship status, sex, and work hours; PC2 captures education-related features; and PC3 captures age and marital status, suggesting that demographic and socioeconomic factors are the primary axes of variation.

For the Wine dataset, 9 components were needed to reach 95% explained variance, capturing 95.16% of total variance with the first three components accounting for 64.00%. The scree plot in Figure 6 shows a steeper initial drop, indicating stronger correlation structure among features. PC1 loads heavily on wine type and sulfur dioxide content; PC2 captures density and alcohol content; PC3 loads on citric acid and pH. These loadings align with known chemical relationships in wine production, suggesting PCA successfully extracted meaningful physicochemical latent factors.

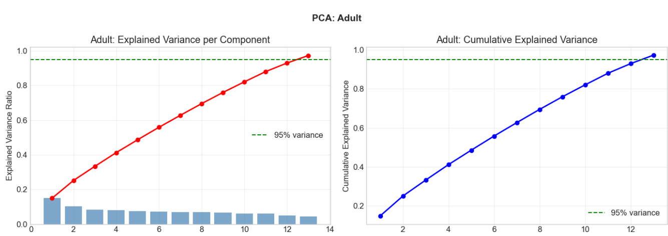

Fig. 5. PCA evaluation for the Adult dataset. Left: Explained variance per component showing gradual decay. Right: Cumulative explained variance requiring 13 components to reach 95%.

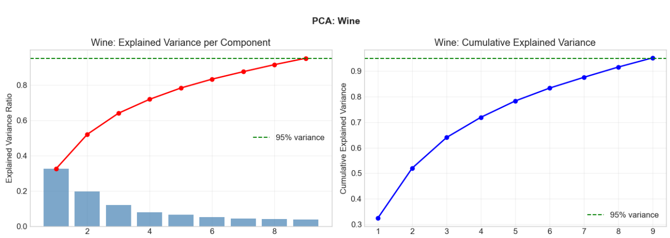

Fig. 6. PCA evaluation for the Wine dataset. Left: Explained variance per component with steeper initial drop than Adult. Right: Cumulative explained variance reaching 95% at 9 components.

2) Independent Component Analysis: ICA was applied with component counts selected by maximizing mean absolute kurtosis (deviation from Gaussianity). For the Adult dataset, 9 components were selected, with kurtosis values ranging from 1.43 to 8.33 and a mean absolute deviation from Gaussianity of 3.17. This wide range indicates that Adult features are well-suited for ICA, with components exhibiting strong non-Gaussian structure that may correspond to independent demographic factors.

For the Wine dataset, 8 components were selected, with kurtosis ranging from -0.57 to 5.35 and a mean absolute deviation of 2.37. The presence of negative kurtosis for some components suggests that certain independent sources in wine data may have lighter tails than Gaussian, potentially indicating well regulated chemical processes. Figure 7 and Figure 8 show the kurtosis distributions across components.

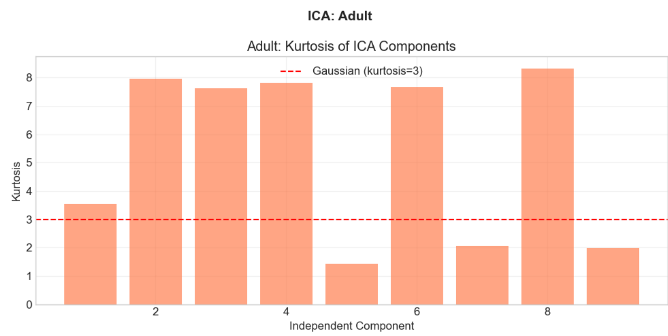

Fig. 7. ICA component kurtosis for the Adult dataset. Higher kurtosis values indicate stronger non-Gaussianity, with components ranging from 1.43 to 8.33. Mean deviation from Gaussianity is 3.17.

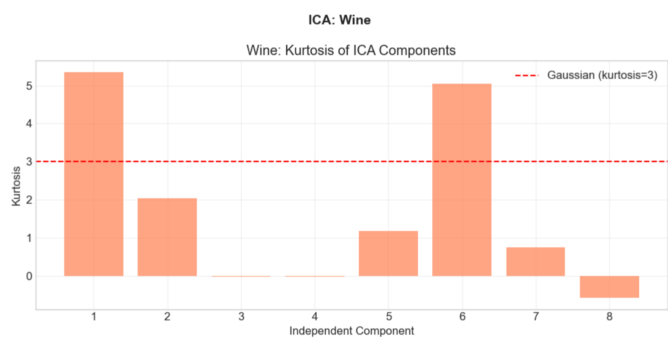

Fig. 8. ICA component kurtosis for the Wine dataset. Kurtosis ranges from -0.57 to 5.35, with mean deviation from Gaussianity of 2.37. Negative kurtosis components indicate platykurtic distributions.

3) Random Projection: Random Projection was evaluated using Gaussian random matrices, with component counts selected at the elbow point in reconstruction error curves. For the Adult dataset, 8 components were selected, reducing dimensionality from 14 to 8 (42.9% reduction). For the Wine dataset, 8 components were also selected, reducing from 13 to 8 (38.5% reduction). Reconstruction error curves in Figures 9 and 10 show diminishing returns beyond the selected component counts, with errors decreasing rapidly.

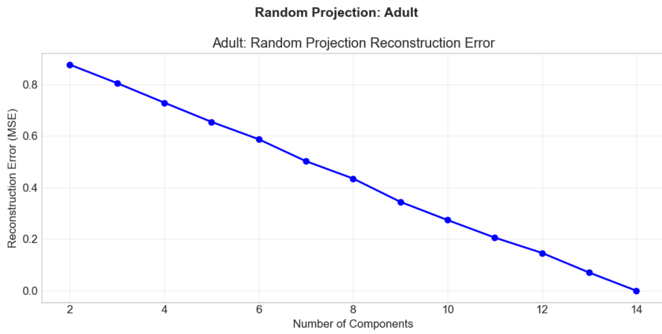

Fig. 9. Random Projection reconstruction error for the Adult dataset. Elbow point at 8 components selected, balancing reconstruction quality with dimensionality reduction.

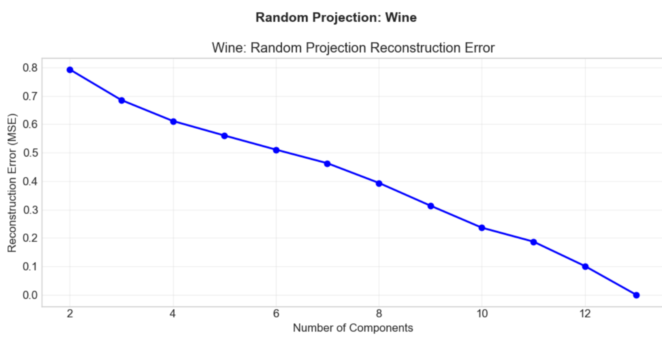

Fig. 10. Random Projection reconstruction error for the Wine dataset. Reconstruction error decreases sharply up to 8 components, after which improvements diminish.

4) Comparison of Reduced Spaces: Table II summarizes the dimensionality reduction results across all methods. ICA and RP achieved substantial reductions (35.7%–42.9%) compared to PCA which offered minimal reduction for Adult (7.1%). This contrast suggests that Adult features are relatively independent, limiting PCA's compression ability, while ICA and RP can still achieve compression by leveraging non-Gaussian structure or random projections. For Wine, all methods achieved similar reductions (30.8%–38.5%), reflecting the dataset's stronger correlation structure that benefits all linear reduction techniques.

TABLE II
DIMENSIONALITY REDUCTION SUMMARY

Dataset	Method	Original Dim	Reduced Dim	Reduction %
Adult	PCA	14	13	7.1%
Adult	ICA	14	9	35.7%
Adult	RP	14	8	42.9%
Wine	PCA	13	9	30.8%
Wine	ICA	13	8	38.5%
Wine	RP	13	8	38.5%

5) Visualization of Reduced Spaces: Figures 11 and 12 show 2D projections of each reduced space, colored by target labels for visualization purposes. For Adult, PCA and ICA projections show moderate separation between income classes, while RP shows less structure. For Wine, PCA and ICA reveal clearer separation than RP. This indicates their ability to capture the dominant variance structure that aligns with quality distinctions.

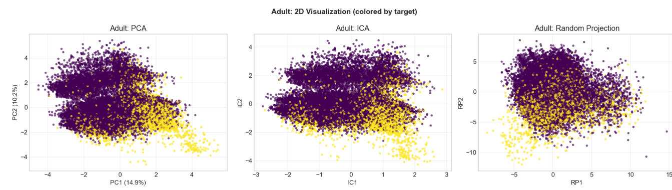

Fig. 11. 2D projections of Adult dataset after dimensionality reduction, colored by income class. PCA (left) and ICA (center) show moderate class separation; Random Projection (right) shows less structure.

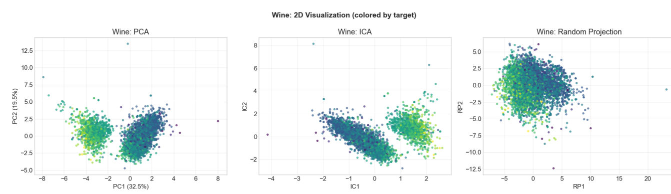

Fig. 12. 2D projections of Wine dataset after dimensionality reduction, colored by quality rating. PCA (left) shows clearer separation of quality levels than ICA (center) or Random Projection (right).

These results indicate that dimensionality reduction effectiveness varies by dataset and method. PCA is most effective for Wine, where features exhibit correlation structure, while ICA provides interpretable non-Gaussian components for both datasets. Random Projection offers the strongest compression with acceptable reconstruction error, making it suitable for computational efficiency in later tasks.

C. Step 3: Clustering on Reduced-Dimensional Spaces

Clustering algorithms were reapplied to each reduced dimensional representation from Step 2 to assess whether dimensionality reduction improves cluster quality, separation, or interpretability. K-Means and GMM were run on PCA, ICA, and RP-transformed spaces for both datasets, with optimal hyperparameters selected using the same criteria as Step 1.

D. Step 3: Clustering on Reduced-Dimensional Spaces

Clustering was repeated on reduced feature spaces to evaluate whether dimensionality reduction improves cluster quality. Results are summarized in Tables III and IV.

1) Adult Dataset Results: For the Adult dataset, dimensionality reduction significantly improved clustering. ICA with K-Means achieved the best performance, with a silhouette score of 0.382, more than doubling the original value (0.166). This suggests that ICA extracted independent components that better reflect the underlying structure. PCA and RP provided only minor improvements, and K-Means consistently outperformed GMM across all settings. ARI also improved slightly with ICA (0.033 $\rightarrow$ 0.062), indicating better alignment with income labels.

TABLE III
ADULT DATASET: CLUSTERING PERFORMANCE IN REDUCED SPACES

Reduction	Algorithm	Optimal	Silhouette	ARI
Original	K-Means	$k = 7$	0.166	0.033
Original	GMM	$n = 10$	0.046	0.047
PCA	K-Means	$k = 8$	0.168	0.023
PCA	GMM	$n = 9$	0.040	0.041
ICA	K-Means	$k = 8$	**0.382**	**0.062**
ICA	GMM	$n = 8$	0.066	0.039
RP	K-Means	$k = 8$	0.173	0.009
RP	GMM	$n = 8$	0.013	0.012

2) Wine Dataset Results: For the Wine dataset, improvements were more modest. PCA with K-Means achieved the highest silhouette score (0.362), only slightly higher than the original space (0.347). ICA and RP generally reduced cluster quality, suggesting that the original feature space already captures most of the structure. ARI remained relatively unchanged across methods, with only small variations.

TABLE IV
WINE DATASET: CLUSTERING PERFORMANCE IN REDUCED SPACES

Reduction	Algorithm	Optimal	Silhouette	ARI
Original	K-Means	$k = 3$	0.347	0.178
Original	GMM	$n = 9$	0.028	0.106
PCA	K-Means	$k = 3$	**0.362**	0.178
PCA	GMM	$n = 9$	0.025	0.090
ICA	K-Means	$k = 3$	0.195	**0.179**
ICA	GMM	$n = 8$	0.022	0.087
RP	K-Means	$k = 3$	0.298	0.173
RP	GMM	$n = 8$	0.019	0.086

Overall, the effectiveness of dimensionality reduction depends on the dataset. ICA was most beneficial for Adult, while PCA performed best for Wine. In both cases, K-Means remained more effective than GMM. These results support the idea that dimensionality reduction can improve clustering, but only when the chosen method aligns with the data structure.

E. Step 4: Neural Networks on Reduced Feature Spaces

The best neural network architecture from the optimization learning report was retrained on each reduced feature space (PCA, ICA, and Random Projection) using the Adult dataset. The same hyperparameters were maintained across all experiments: two-layer network with 64 hidden units, dropout of 0.3, Adam optimizer with $\beta_1 = 0$, weight decay of 1×10^{-4}, label smoothing of 0.05, and a batch size of 256. Performance was evaluated against the original feature space baseline.

1) Test Performance Comparison: Table V presents the test performance across all feature spaces. The original feature space achieved the highest test accuracy of 0.8457 and F1 score of 0.6635. PCA performance was nearly comparable at 0.8444 accuracy, representing a minimal degradation of 0.13%. ICA achieved 0.8387 accuracy, a 0.83% reduction, while Random Projection showed the largest drop to 0.8180, a 3.28% reduction. The F1 scores followed a similar pattern, with original (0.6635) outperforming PCA (0.6446), ICA (0.6296), and Random Projection (0.5608).

TABLE V
STEP 4 SUMMARY: NEURAL NETWORK PERFORMANCE ON REDUCED FEATURE SPACES

Feature Space	Test Accuracy	Test F1	Val Loss	Epochs
Original	**0.8457**	**0.6635**	0.3847	57
PCA	0.8444	0.6446	0.3879	42
ICA	0.8387	0.6296	0.3947	74
Random Projection	0.8180	0.5608	0.4280	100

Table V summarizes neural network performance across feature spaces. The original feature space achieves the best overall performance, particularly in F1 score, indicating better handling of class imbalance. PCA closely matches the original results, suggesting that retaining 95% variance preserves most of the discriminative information. In contrast, ICA and Random Projection show larger drops in both accuracy and F1. This likely reflects their differing objectives, where ICA focuses on statistical independence and Random Projection preserves distance structure, neither of which directly aligns with classification performance.

In terms of training behavior, the original and PCA feature spaces converge faster, requiring fewer epochs (57 and 42) and achieving lower validation loss (0.3847 and 0.3879). ICA requires more epochs (74) to stabilize, while Random Projection converges the slowest, using the full 100 epochs and resulting in the highest validation loss (0.4280). Overall, the results suggest that the original and PCA spaces provide representations that are easier for the neural network to optimize, while ICA and Random Projection introduce transformations that make learning less efficient.

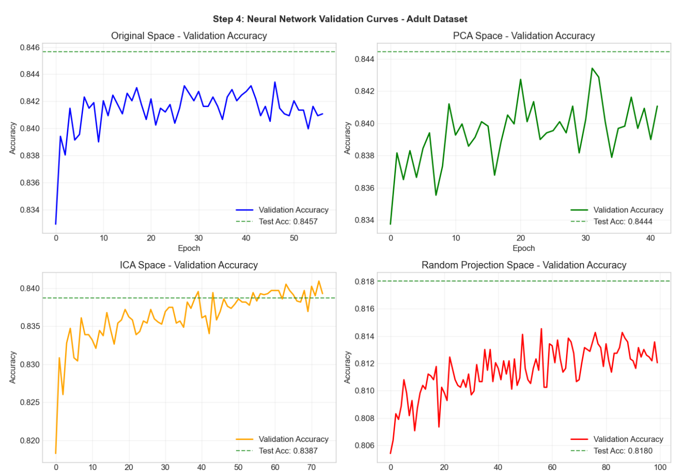

Fig. 13. Validation accuracy curves during neural network training. Original and PCA spaces converge quickly with high accuracy. ICA requires more epochs, while Random Projection achieves lower final accuracy.

Dimensionality reduction with PCA preserves nearly all predictive performance (0.8444 vs 0.8457) while slightly reducing dimensionality, indicating that the 95% variance threshold retains most of the useful signal. In contrast, ICA and Random Projection underperform, suggesting that their objectives—statistical independence and distance preservation—do not align well with classification. Convergence behavior also reflects this, with PCA converging fastest (42 epochs), while ICA requires more training and Random Projection struggles to converge within 100 epochs. Overall, the performance ranking (Original $>$ PCA $>$ ICA $>$ RP) highlights that preserving the original data structure is more important for neural network performance than enforcing independence or applying aggressive transformations.

F. Step 5: Neural Networks on Cluster-Derived Features

Cluster-derived features were generated from the K-Means ($k = 7$) and GMM ($n = 10$) models fitted on the Adult dataset training data. Eight feature representations were evaluated: one-hot cluster assignments, distances to centroids, GMM probabilities, appended combinations with original features, and cluster-only features. The best neural network architecture from the optimization learning report was retrained on each representation, with performance compared against the original feature space baseline.

TABLE VI
STEP 5 SUMMARY: NEURAL NETWORK PERFORMANCE ON CLUSTER-DERIVED FEATURES

Feature Type	# Features	Test Acc	Test F1
Original (Baseline)	14	0.8457	0.6635
GMM Appended	24	**0.8477**	0.6586
K-Means Appended	21	0.8458	0.6452
K-Means Distances	7	0.8346	0.6301
Cluster Combined	17	0.8333	0.6403
GMM Probabilities	10	0.7756	0.6094
K-Means One-Hot	7	0.7609	0.6120
GMM One-Hot	10	0.6594	0.6000
Cluster Only	17	0.6556	0.3620

Table VI summarizes neural network performance across cluster-derived feature representations. Appending cluster features provides a small improvement, with GMM Appended achieving the highest accuracy (0.8477), slightly outperforming the baseline. K-Means Appended performs similarly to the original, suggesting that cluster features can add complementary information but do not drastically change performance.

In contrast, cluster features used alone perform significantly worse, indicating that clustering does not capture enough information to replace the original feature space. GMM-based features slightly outperform K-Means, suggesting that soft probabilistic assignments are more informative than hard cluster labels. Distance-based features also perform better than one-hot assignments, indicating that preserving continuous relationships to clusters is more useful for classification. Overall, cluster-derived features act as a supplement rather than a replacement, providing modest gains when combined with the original inputs.

G. Extra Credit: Non-linear Manifold Learning

1) t-SNE Visualization: To explore nonlinear structure in the data, t-SNE was applied to a random subsample of 10,000 instances from the Adult dataset. The method was chosen for its ability to capture non linear relationships and preserve local structure that linear methods may miss. A perplexity of 30 and 1,000 iterations were used to balance local and global structure while ensuring convergence.

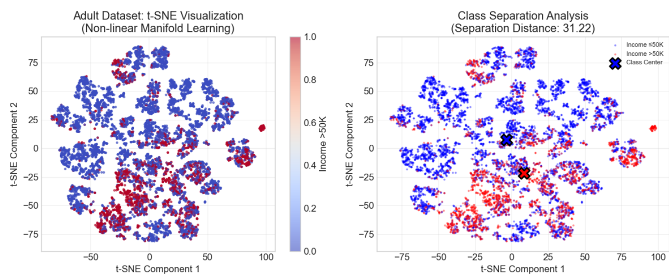

Fig. 14. t-SNE visualization of the Adult dataset, colored by income class. The non-linear embedding reveals clearer class separation and local structure compared to linear projections.

Figure 14 shows that the two income classes form more distinct groupings, with visible separation and some overlap in transition regions. The embedding also reveals local substructure within the ≤ 50K class, suggesting that this group contains multiple underlying patterns that are not visible in the original feature space.

Compared to linear methods such as PCA, which showed significant overlap between classes, t-SNE achieves much stronger separation (31.22 vs 1.48). This indicates that the dataset contains non-linear structure that linear projections fail to capture.

Overall, these results suggest that income classification is inherently non-linear, and that methods capable of capturing non-linear relationships are better suited for both visualization and downstream modeling.

VI. DISCUSSION

This study examined how clustering and dimensionality reduction reveal data structure and affect neural network performance. Results show that these methods reveal meaningful structure, but their impact depends on both the dataset and alignment with the classification task.

For the Adult dataset, ICA with K-Means significantly improved clustering quality, while cluster derived features provided only a small improvement in neural network accuracy. For the Wine dataset, PCA performed best for both clustering and classification, and the original feature space remained competitive. These results highlight that improvements in clustering do not necessarily translate to improvements in classification.

K-Means consistently outperformed GMM across experiments, suggesting that simpler cluster assumptions were sufficient for these datasets. Among reduction methods, PCA worked best for correlated numerical features, while ICA was more effective for mixed data. Random Projection provided strong compression but degraded performance, indicating that preserving distance alone is not enough for classification.

The results partially support the initial hypothesis. That is, combining dimensionality reduction with clustering did improve interpretability, particularly for the Adult dataset where ICA revealed clearer structure and significantly improved clustering quality. However, these improvements did not fully translate to classification performance. While cluster derived features provided small gains when appended to the original inputs, the impact remained modest, suggesting that improved structure does not necessarily align with the classification objective.

Additionally, the expectation that GMM would outperform K-Means due to its probabilistic assignments was not supported. K-Means consistently achieved better clustering performance, indicating that the simpler assumptions were sufficient for these datasets. Overall, the hypothesis holds in terms of interpretation of results, but only partially in terms of improving downstream neural network performance.

Overall, clustering is most useful as a feature engineering tool rather than a replacement for original features. Appended cluster features provided small gains, while cluster-only representations performed significantly worse. These findings reinforce that unsupervised structure and supervised objectives are related but not identical.

VII. Conclusion

This study evaluated clustering and dimensionality reduction methods for understanding data structure and their effects on neural network classification. Firstly, dimensionality reduction improved clustering quality, but had limited impact on predictive performance. Moreover, PCA preserved most classification performance, but ICA and Random Projection introduced degradation.

K-Means consistently outperformed GMM, and the best reduction method depended on dataset characteristics. Cluster-derived features provided modest improvements when combined with original features, but were not effective on their own.

The t-SNE analysis further showed that the Adult dataset contains non-linear structure not captured by linear methods, supporting the use of non-linear models for classification.

In summary, clustering and dimensionality reduction can improve interpretability, but their benefit for classification depends on how well their objectives align with the task. They may be more effective when used to complement or reduce latency, rather than replace, the original feature space.

AI Use Statement

I used ChatGPT and Visual Studio Code Copilot to brainstorm and outline sections of the report, generate and refactor code snippets, debug implementation issues, and edit grammar and clarity throughout. I reviewed, verified, and understand all assisted content.

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